

# Statistical & Theoretical Assumptions

## BrainKAN Explainability Framework

This document explicitly details the mathematical and statistical assumptions underlying the BrainKAN Explainability framework. Understanding these boundaries is critical for interpreting the  $Z$ -score and avoiding causal overclaims.

### Assumption 1: Exchangeability of Label Permutation

**Statement:** Under the null hypothesis  $H_0$  (no task-associated nonlinear structure), the distribution of the observed geometric test statistic  $G_e(f, X)$  is invariant to the permutation of task labels  $Y$ .

$$(X_i, Y_i)_{i=1}^n \stackrel{d}{=} (X_i, \pi(Y)_i)_{i=1}^n \quad (1)$$

**Why it is reasonable:** The BrainKAN model minimizes cross-entropy against  $Y$ . If  $Y$  contains no genuine task information linked to  $X$ , any geometric deformation learned by the model is purely a product of model flexibility, initialization, and input data structure.

**Violation Consequence:** If the permutation strategy breaks inherent dependencies (e.g., temporal autocorrelation in fMRI or fold-level pseudo-replication) that the null model is supposed to preserve, the empirical null variance  $\sigma(G_{null})$  will be underestimated, leading to **inflated false positives** and artificially high  $Z$ -scores.

### Assumption 2: Null Preserves Nuisance Geometry

**Statement:** The label-shuffled null model  $f_{null}$  successfully captures the geometric deformation induced by reference domain shifts  $D(X, f)$  and baseline model capacity, independently of the biological signal  $S(f)$ .

$$G(f_{real}, X) = S(f) + D(X, f_{real}) \quad (2)$$

$$G(f_{null}, X) = D(X, f_{null}) \quad (3)$$

**Why it is reasonable:** Deep neural networks exhibit implicit bias and expressivity even when fitting noise. By training  $f_{null}$  on the exact same input distribution  $X$  with the same hyperparameter budget, it absorbs the geometric complexity necessary to route the data distribution, minus the task constraint.

**Violation Consequence:** If the true signal  $S(f)$  is deeply entangled with the reference deformation (i.e.,  $D$  is strongly multiplicative rather than additive), the subtraction  $G_{real} - \mu(G_{null})$  will fail to fully isolate the signal. This leads to **biased calibration** where apparent nonlinearities are artifacts of input density rather than function mechanism.

### Assumption 3: Reference Grid Approximates Operating Domain

**Statement:** The fixed reference domain  $X_{ref}$  constructed from the source cohort adequately covers the regions of high curvature and function variation expected in the target cohort.

**Why it is reasonable:** The framework mandates calculating the Support Overlap  $O_j$  to explicitly quantify this overlap. If  $O_j \rightarrow 1$ , interpolation holds.

**Violation Consequence:** If the target cohort operates in a completely disjoint input regime ( $O_j \ll 1$ ), spline evaluations become zero-order extrapolations. In this regime, numerical derivatives explode or vanish arbitrarily. The computed geometry  $G_e$  becomes mathematically undefined with respect to the true biological function, rendering the identifiability assessment **invalid**.